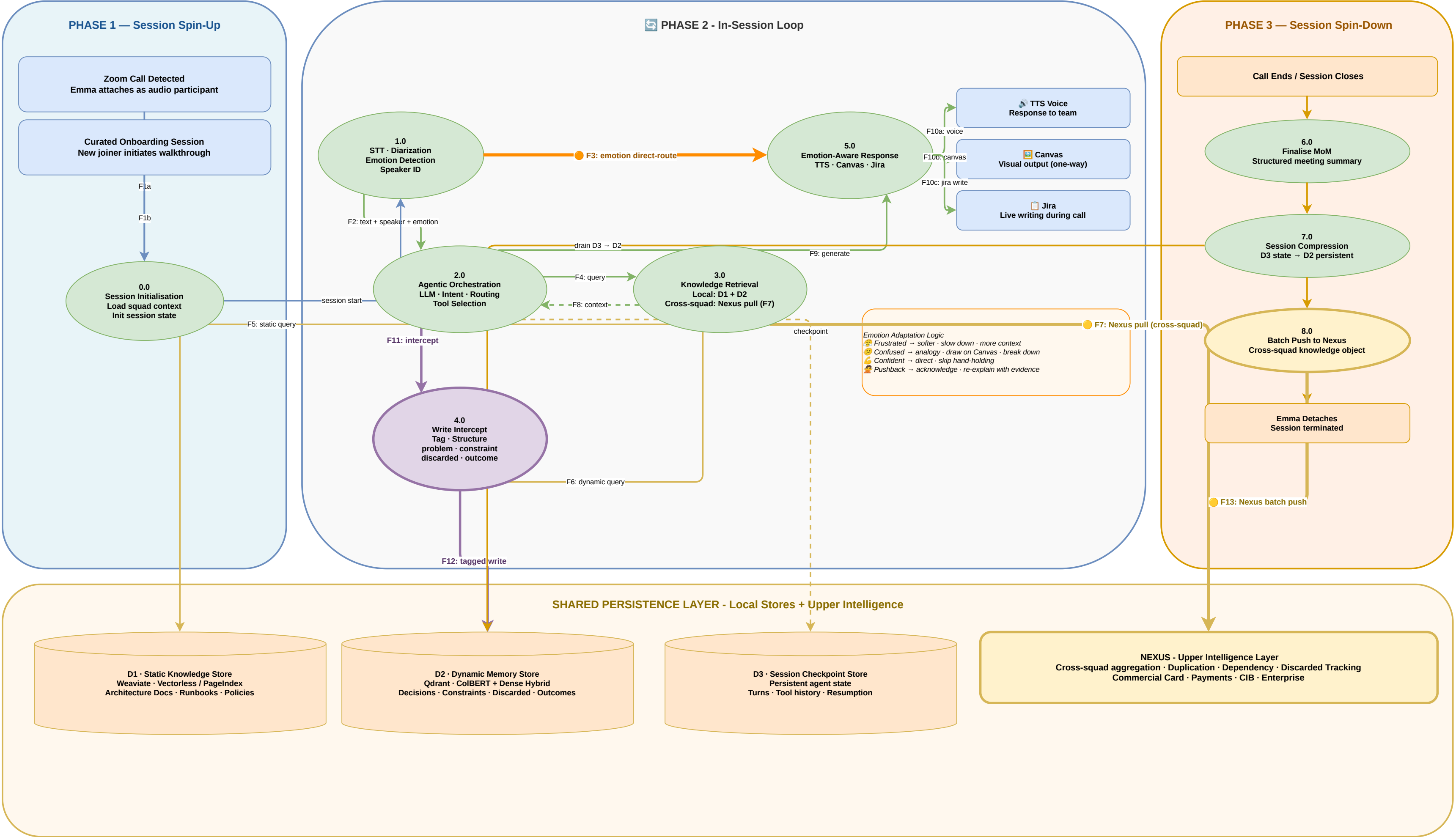


Emma — Squad Session Lifecycle Architecture

Solid = data flow · Dashed = async / lifecycle · ● Emotion path · ● Write Intercept · ● Bidirectional Nexus



Emma Platform — Institutional Memory Architecture

Knowledge flows UP (solid arrows) · Intelligence pushes DOWN (dashed arrows) · Every sub-LOB has its own Nexus

ENTERPRISE NEXUS

Frontier LLM · Enterprise Vector DB (Qdrant / Pinecone)
Rolls up ALL LOBs: CIB · CCB · AWM · CT

- Cross-LOB pattern detection · Org-wide duplication identification
- Discarded decision tracking + constraint resolution monitoring
- Pushes resolved constraints back down to all LOB Nexuses
- Strategic dependency mapping · Systemic inefficiency surfacing

↑ Batch pull from ALL LOB Nexuses (sprint cadence)
↓ Event-driven push back down to LOBs and Emmas
↔ Emma pulls from Nexus on-demand mid-meeting (no direct Emma-to-Emma)

Alerts to: Architecture Leadership · Impacted PO

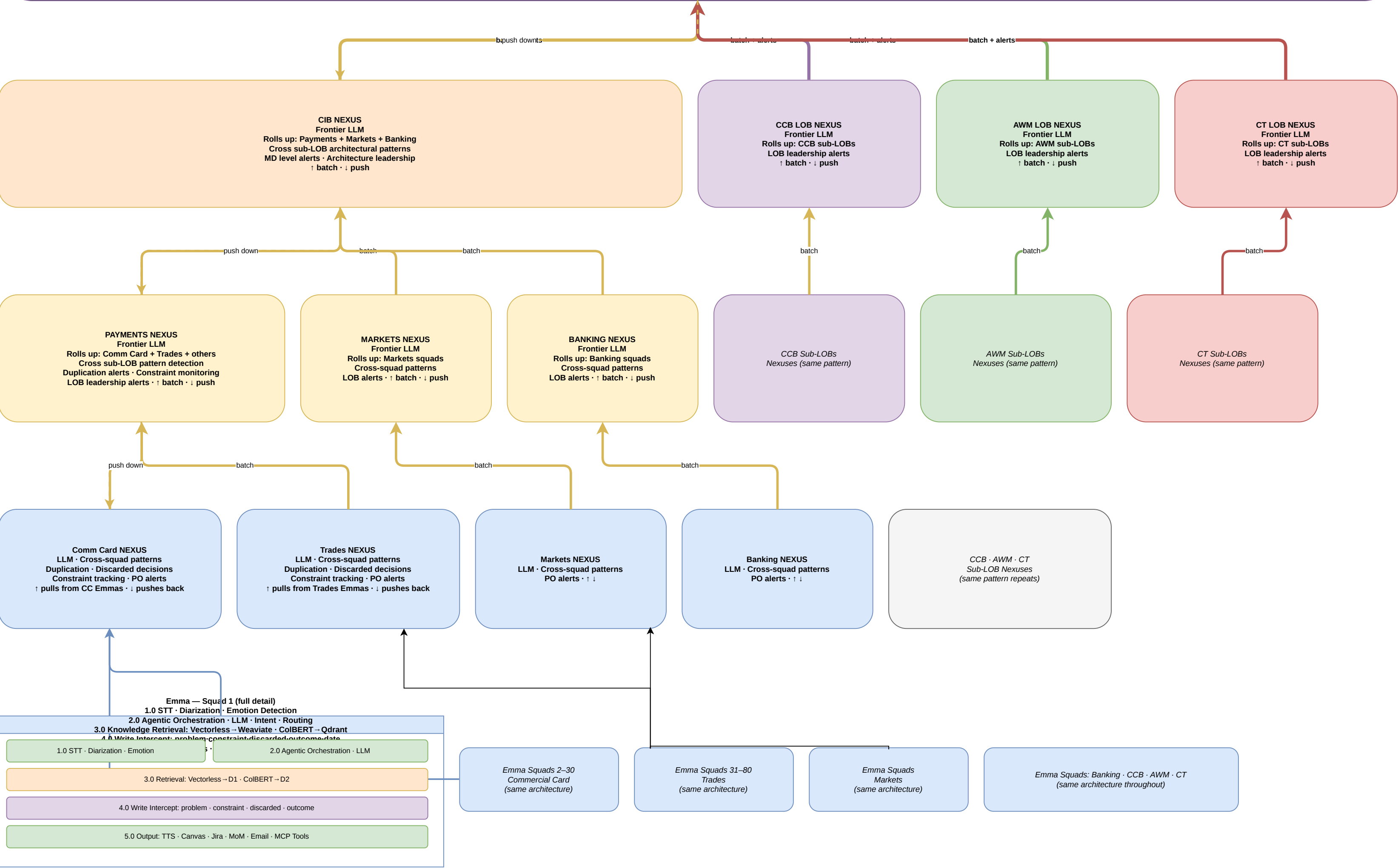
THE PLAYBOOK

Same pattern at every level:
Emmas → sub-LOB Nexus
→ LOB Nexus → Enterprise

Sonnet LLM at sub-LOB
Frontier LLM at LOB+

Knowledge UP (batch)
Intelligence DOWN (event)

Solid arrows = Knowledge UP (batch)
Dashed arrows = Intelligence DOWN (event-driven)
↔ Emma pulls from Nexus on-demand mid-meeting
See Diagrams 3 & 4: how Nexus works



📖 HOW NEXUS WORKS — Batch Intelligence at the Commercial Card Level

Every sprint cycle, Nexus runs in **batch mode** across the cross-squad vector databases — accumulated from each squad Emma's session-end push (🟡 F13). No meetings required. No questions asked. Pure pattern matching over what every team is building, debating, and discarding.

Two things Nexus surfaces automatically:

⚠️ SCENARIO A — Duplication caught early

Team B's PO drops "phone number validation" into next sprint. Nexus reads the vector DBs and knows Team A shipped phone validation 2 sprints ago — already in production. Alert fires to both POs: **"Team B is about to rebuild what Team A already shipped. Align before sprint commit."**

⚠️ SCENARIO B — Dependency surfaced before it bites

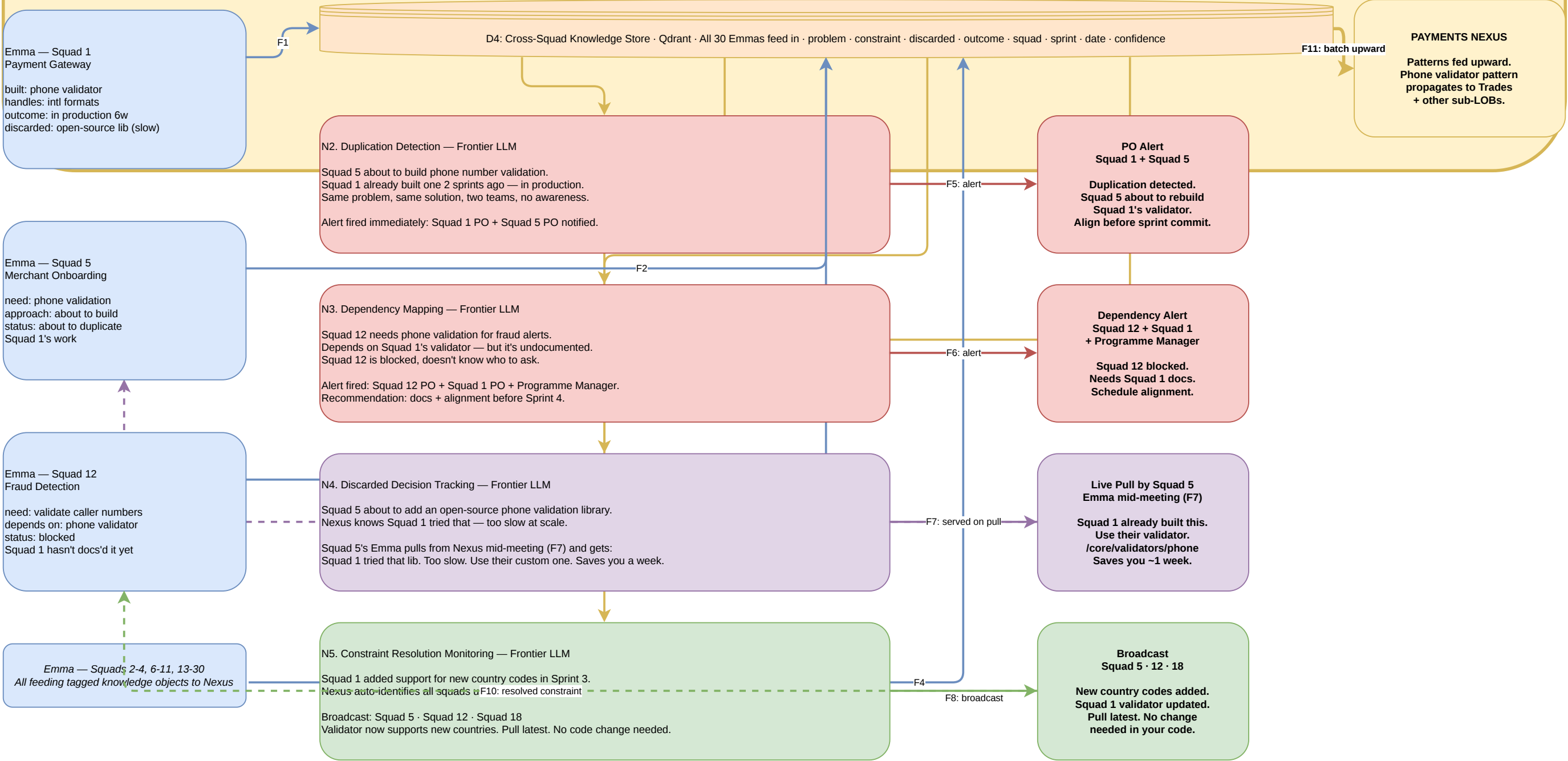
Team B is building merchant onboarding. Nexus subtly detects: this feature will need Team A's payment validator downstream — but Team A hasn't published the API contract yet. Alert fires to both POs: **"Team B has a hidden dependency on Team A's work. Align now, not in Sprint 4 when integration breaks."**

Both interventions happen **BEFORE damage is done**. Multiply by 30 squads, 4 LOBs, every sprint cycle.

Bidirectional Nexus (🟡 F13 session push feeds data in + 🟡 F7 live query serves it back) is the mechanism. The value lives in between — batch pattern detection that no individual team can see.

Commercial Card NEXUS — Cross-Squad Intelligence

How Nexus detects duplication, dependency and discarded decisions across squads — before damage is done



Nexus — How It Works Internally

Batch + on-demand query architecture · same pattern repeats at every level (Commercial Card → Payments → CIB → Enterprise)

